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Manufacturing and Testing a Carbon Fiber Reinforced Epoxy I Beam

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Abstract

This report presents the design, manufacturing, and testing of a carbon fiber reinforced epoxy I-beam developed for the 2026 SAMPE bridge competition. The beam was designed using Classical Laminate Theory (CLT) in combination with Timoshenko beam theory to estimate laminate properties, global beam stiffness, midspan deflection, and first-ply-failure behavior under three-point bending. The final design used $[0]_6$ flange reinforcement and $[\pm 45]_3$ s web facings around a Nomex honeycomb core. Based on the design calculations, the beam was predicted to weigh 348 g, achieve an effective flexural modulus of 45.95 GPa, and reach first-ply failure at approximately 7700 lb with a corresponding midspan displacement of 4.45 mm. The predicted governing failure mode was compressive failure in the upper flange, while honeycomb shear was recognized as a potential weak point not fully captured by the laminate-level model.

The manufacturing plan originally called for fabrication using prepreg laminates cured in a custom steel C-channel mold inside an autoclave, with the web stabilized by two layers of Nomex honeycomb. During fabrication, the initial build revealed major manufacturing problems. In particular, the use of expanding foam in the core produced an uneven bonding surface and introduced internal stresses that contributed to poor laminate bonding and flange warpage during cure. A second build was therefore produced using adhesive film instead of foam and improved mechanical constraint of the flanges during curing. This revised approach resulted in a more stable beam geometry and a successfully testable specimen.

The final beam weighed 432.4 g and measured $24 \times 3 \times 2.8$ in, with an average flange thickness of 0.08 in and a web thickness of 0.48 in. Three-point bending testing showed an experimentally observed peak load of 2,376 lbf, substantially lower than the analytically predicted failure load. In addition, the first physical failure was observed in the web region rather than as upper-flange compressive failure. The measured response indicates that the beam did not fail according to ideal laminate strength predictions alone, but instead through a combination of core weakness, imperfect bonding, stress concentrations, interlaminar effects, and manufacturing-induced geometric imperfections.

Overall, the results show that the analytical model was useful for establishing laminate architecture, estimating stiffness trends, and identifying likely load paths, but it over-predicted structural performance because it assumed ideal bonding, ideal geometry, and simplified stress states. The discrepancy between predicted and observed behavior highlights the dominant role of fabrication quality, interface integrity, and cure constraints in determining the actual performance of thin-walled composite structures. This project therefore demonstrates that successful composite beam design requires close integration of analysis, manufacturing planning, tooling design, and experimental validation.

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1 Manufacture and Test Plan (Assignment 1)

1.1 Section Abstract

This section describes the design and manufacturing process for a carbon fiber reinforced polymer I-beam eligible for the 2026 SAMPE composite bridge contest. The static 3-point bending response of the beam is predicted using a combination of classical laminate theory and the Timoshenko-Ehrenfest beam theory. Results are used to determine first-ply-failure and approximate critical buckling stress in the web. Since the model assumes perfect layup, a sensitivity analysis is performed to determine the effect of manufacturing error on the final behavior of the beam. The processing steps to fabricate the I-beam are discussed as well as alternative manufacturing approaches. Analysis of tooling is performed to assess material heating during the cure process.

1.2 Design Objective

The objective of this project is to design and fabricate a composite beam that meets the requirements described in sections 2(b) and 3.2 in the 2025 SAMPE composite bridge contest rule book for carbon/aramid fiber reinforced I-beams (see ref. [1]). The bridge will be loaded up to 7200 lbf (32 kN) in 3-point bending with a span of 23 in (584 mm). Based on data from previous competitions, the weight goal for this bridge is 600 g. The bridge must withstand at least 1500 lbf without catastrophic failure of any laminate sections.

1.3 Design Introduction

1.3.1 Composite Laminate Bending Mechanics

To meet the stated design objectives, an analytical approach is used to model the composite structure. Classical laminate theory (CLT) is a key design tool for analyzing the response of multi-layer composite plates to specified in-plane loads. Each layer (ply) has its own stiffness matrix corresponding to its fiber orientation. Each stiffness matrix (Q_{ij}^k for the k th lamina) is weighted according to the distances (h_k and h_{k-1}) of the ply surfaces to the mid-plane and summed to obtain the 3 x 3 **A**, **B**, and **D** sub-matrices of the ABD matrix. The summations are shown in Equations 1 through 3.

$$\mathbf{A} = \sum_{k=1}^n Q_{ij}^k (h_k - h_{k-1}) \quad (1)$$

$$\mathbf{B} = \sum_{k=1}^n \frac{1}{2} Q_{ij}^k (h_k^2 - h_{k-1}^2) \quad (2)$$

$$\mathbf{D} = \sum_{k=1}^n \frac{1}{3} Q_{ij}^k (h_k^3 - h_{k-1}^3) \quad (3)$$

Note that the h values for lamina surfaces below the midplane are negative, while those above the midplane are positive. This is mostly relevant for $\mathbf{A}$ terms and $\mathbf{D}$ terms because $\mathbf{B}$ includes only h^2 terms which are always positive. The fiber orientation of a continuous fiber-reinforced lamina is measured from global coordinates x and y that are consistent for the entire laminate. Applied stresses, moments, and mid-plane strains and curvatures are also measured from global coordinates. The relationship between the applied forces and moments ($N_x, N_y, N_{xy}, M_x, M_y, M_{xy}$) and the resulting strains and curvatures ($\varepsilon_x^o, \varepsilon_y^o, \gamma_{xy}^o, \kappa_x, \kappa_y, \kappa_{xy}$) is given by Equation 4.

$$\begin{pmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{pmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{pmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_{xy}^o \\ \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{pmatrix} \quad (4)$$

For a composite laminate beam in pure bending, as shown in Figure 1, the only nonzero component of the load tensor is M_x . The terms $B_{11}, B_{12}, B_{16}, D_{12}, D_{11}$, and D_{16} are coupling terms. Nonzero values of B_{11} and B_{12} produce extensional strains from an applied bending moment. Similarly, nonzero B_{16} and D_{16} produce shear and twisting strains respectively. The D_{11} term couples bending in the x -direction and the D_{12} term couples curvature in the y -direction to bending in the x -direction. The overall relation between M_x and the ABD matrix terms is shown in Equation 5

$$M_x = B_{11}\varepsilon_x^o + B_{12}\varepsilon_y^o + B_{16}\gamma_{xy}^o + D_{11}\kappa_x + D_{12}\kappa_y + D_{16}\kappa_{xy} \quad (5)$$

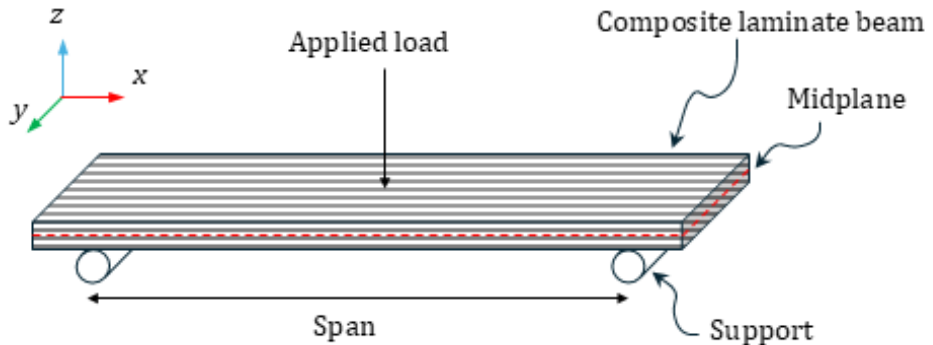


Figure 1: Flat composite laminate 3-point bending. The coordinate system shows the orientation of the beam with respect to each axis.

Symmetry of plies about the mid-plane can prevent unwanted strains from arising when load is applied. For example, if the laminate is a balanced, symmetric cross-ply (there are an equal number of 0° and 90° plies, and they are symmetric about the mid plane), all $\mathbf{B}$ and $\mathbf{A}$ as well as the D_{16} and D_{26} terms become zero. This eliminates the coupling between x-axis bending and extensional strain. Coupling between x-axis bending and y-axis curvature and twisting are also removed. The resulting strain due to the applied load is therefore expected to be predominantly x-axis curvature.

Failure of continuous fiber-reinforced polymer composites is complex. Laminate failure modes include fiber breakage, fiber pull-out, fiber buckling, matrix cracking, and delamination. The laminate strength is considered by assessing which ply will fail first, second, third, and so on until the part is fully ruptured. On a ply-by-ply basis, if the fiber-matrix interface is sufficiently strong, the component (fiber or matrix) with the lowest ultimate strain will fail first.

Beyond standard unidirectional-ply laminates, lightweight composite beams with various unique architectures can be designed to withstand bending stresses. Common architectures include I-beams, rectangular tubes, and sandwich panels.

1.3.2 Sandwich Panels

Sandwich panels are constructed from low-density core material between two thin, high-strength facings. The core of a sandwich beam in bending reacts out-of-plane shear and compressive stresses while the facings primarily react in-plane compressive and tensile load. Assuming the laminate facings are much thinner and stiffer than the core, the bending stiffness of a symmetric sandwich panel scales with the square of the core thickness h_c [2]. The relationship is shown in Equation 6

$$D_{ij} \approx \frac{1}{2} Q_{ij,f} h_f h_c^2 \quad (6)$$

Common core materials for sandwich composites in the aerospace industry are polymer foams and aluminum alloy or Nomex honeycombs. Since core materials are very low-density—aerospace foam core, for example, can be as light as 0.0058 lb/in^3 (0.16 g/cm^3)—a thick core can considerably improve bending stiffness with minimal added weight. Sandwich panels can be found in aircraft fuselages, control surfaces, bulkheads, and fairings.

Bending failure modes of composite sandwich beams include compressive buckling or kinking of face sheets, delamination of face sheets from the core, and transverse shear failure of the core. In the case of honeycomb sandwich panels, buckling failure is sometimes observed in the vertical hexagonal cells as they experience compressive and transverse shear loading [3].

1.3.3 Laminated Composite Beams

Fiber reinforced polymer (FRP) thin-walled beams with various cross sections are found in applications such as helicopter tail booms and spars in airplanes. Composite I-beams are useful in applications with high bending loads, as the flanges concentrate material at a maximum distance from the neutral axis, providing a high second moment of area with respect to the neutral plane and consequently a high bending stiffness. FRP I-beams are typically manufactured with glass fiber reinforcement, and are not as common as square or round tube beams. This is likely because I-beams are susceptible to cracking at the web-flange intersection and because the open section provides lower torsional stiffness than closed sections.

1.3.4 Examples of Composite Beams

Various approaches to beam fabrication can be found in the literature. Two examples are included here.

Approach 1: Carbon/epoxy I-beam with flexible Nomex honeycomb core [4] [5]. The advantages of the sandwich beam construction can be employed in the design of laminate I-beams. For example, implementing a honeycomb core within the web provides improved buckling stability and flexural rigidity with minimal added weight. Honeycomb core structures offer important advantages over homogeneous cores. Nomex honeycomb core is designed so that its in-plane response is dominated by cell wall bending and rotation rather than simple linear stretching. This concept combines the high specific stiffness of a cellular core with a controllable, strain dependent effective modulus: as curvature increases in flexure, cell walls experience larger rotations and stretching, and the effective in-plane modulus in the loading direction rises substantially, making the I-beam increasingly resistant to further bending without a large mass penalty. Existing work on Nomex honeycomb cores also emphasizes that transverse shear and local cell wall instabilities can limit flexural capacity, so this flexible core concept, with validated nonlinear stiffness predictions, provides a rational way to exploit and control that behavior rather than simply over-designing the core to avoid it.

Approach 2: Laminated composite honeycomb sandwich panels optimized against shear buckling [3]. A second interesting approach is to fabricate the cellular core itself from laminated composite plates, rather than metallic foil, and to tailor the laminate layup in the cell walls to delay shear buckling under flexure-induced transverse shear. In this concept, the face sheets remain relatively conventional fiber reinforced laminates that carry in-plane bending stresses, but the honeycomb walls are made from angle ply composite laminates with bend-twist coupling explicitly accounted for. Recent semi-analytical models treat each wall segment as a laminated plate with rotationally restrained edges and use laminate plate buckling theory to predict the critical shear buckling strain of the core as a function of fiber orientation, stacking sequence, and core geometry. Semi-analytical results show that by selecting appropriate angle ply layups in the cell walls and accounting for bend twist coupling, the predicted critical global transverse shear strain for core buckling can be substantially increased

compared to isotropic cores, directly translating into higher allowable bending loads before local instability occurs. Additionally, because the model relates global shear strain in the sandwich to local shear in each wall segment, it provides a design tool to tailor wall layups such that the onset of shear buckling in the core coincides with or exceeds the desired ultimate flexural capacity of the face sheets, enabling lighter cores without sacrificing flexural strength.

1.4 Design Approach

The aim of this analysis is to predict the static 3-point bending response of a constant-section carbon FRP laminate I-beam using CLT and the Timoshenko beam theory. Geometric constraints provided by the SAMPE student competition require the beam to be at least 24 in (609 mm) long and the cross-section to be under 4 in x 4 in (101.6 mm x 101.6 mm). The design load is 7500 lbf applied at mid-span as shown in Figure 2. A higher test load of 10000 lbf is applied in the analysis, providing an initial safety factor of 1.4. The bridge weight goal is <600 g, and the design must be compatible with available manufacturing tools.

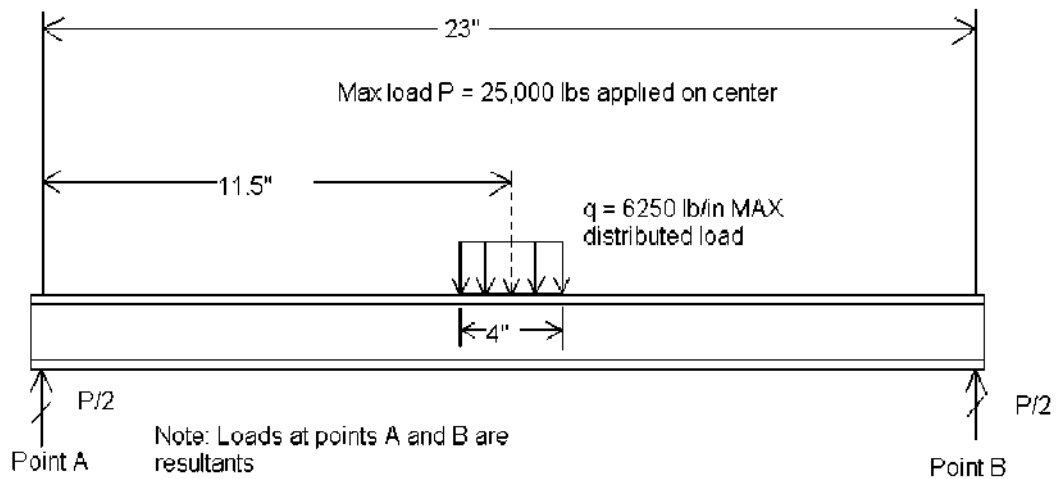


Figure 2: Bridge load case illustration from the composite bridge competition rules [1]

The analysis is done in three stages: ply-level properties, laminate properties, and the behavior of the built-up structure. The I-beam construction requires separate analysis of the flanges and web, as the composite architecture and loading conditions are different for each. The beam web consists of two C-channels connected back-to-back with a honeycomb core between them to provide stability with minimal added weight. Unidirectional plies are added on the top and bottom to form the flanges. This construction is preferred over simply joining together perpendicular flat laminates to create an I section, as it reduces the risk of cracks forming at the intersections between the web and flanges. An important consequence of the C-channel design is that the flanges consist partly of the laminate for the web and partly of the unidirectional top

and bottom layers. This structure is assumed to be continuous in the beam model, though in reality, there would be a discontinuity where the web laminates meet the core, as shown in Figure 3.

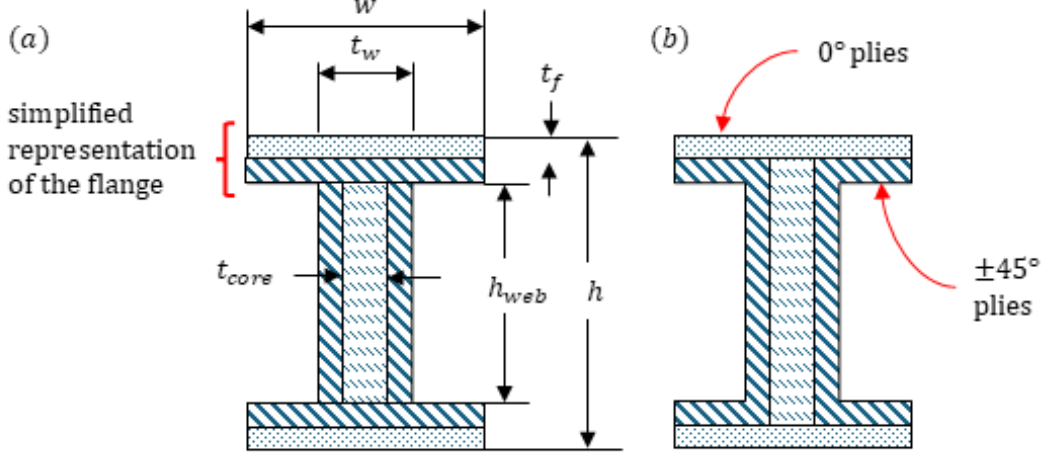


Figure 3: Diagram of the I-beam cross-section used for modeling with CLT. (a) shows the model core and flange configuration with dimensions; (b) shows the more realistic geometry of the core and flanges.

The ply properties are obtained using data from the DA 409/G35 Carbon Epoxy Prepreg System data sheet by APCM [6]. Additional material property resources are cited in Table 1 in the Design Results section. The semi-empirical Halpin-Tsai equation is used to determine transverse tensile modulus based on the fiber volume fraction and individual resin and fiber properties. For the honeycomb core, the present analysis regards the in-plane elastic moduli as constant, though as mentioned in the design introduction, deformation affects the core stiffness (see ref. [4]). The total weight of the bridge is estimated based on the number of plies, dimensions of the structure, and areal density of the prepreg and core materials.

A Python script is used to obtain the stiffness matrices for each ply in the structural coordinates and generate ABD matrices. The I-beam is a multi-part 3-D structure, which means that it does not possess a unified ABD matrix. For this analysis, equivalent engineering constants for the beam are derived from the separate web and flange ABD matrices. Since the flanges are asymmetric and have non-zero coupling matrices, the equivalent tensile (E_x , E_y) and shear (G_{xy}) moduli are determined by multiplying the inverse ABD matrix for each component by a unit load tensor and entering the resulting strain into the appropriate equation of Equations 7 – 9. In these equations, t denotes the thickness of the entire laminate and ε_x , ε_y , and ε_{xy} denote the x, y, and shear strains respectively.

$$E_{x,laminate} = \frac{1}{t\varepsilon_x} \quad (7)$$

$$E_{y,laminate} = \frac{1}{t\varepsilon_y} \quad (8)$$

$$G_{xy,laminate} = \frac{1}{t\gamma_{xy}} \quad (9)$$

The overall web construction including the honeycomb core is symmetric about the xz -plane, therefore the simplified relations 10 – 12 are applied to compute the equivalent shear and elastic moduli. Here, the components of the inverted ABD matrix are denoted by lowercase letters a_{ij} , b_{ij} , and d_{ij} .

$$E_{x,laminate} = \frac{1}{ta_{11}} \quad (10)$$

$$E_{y,laminate} = \frac{1}{ta_{22}} \quad (11)$$

$$G_{xy,laminate} = \frac{1}{ta_{66}} \quad (12)$$

The web laminate lies in the xz -plane, while the moment acts about the y -axis. Consequently, the expected bending mode is incompatible with the current assumptions for CLT. In the analysis, this is resolved by assuming that the bending stresses in the web act as normal and shear stresses in-plane. To determine the equivalent flexural modulus for the entire beam, the web and flange stiffness values are combined using the parallel axis theorem as shown in Equation 13 where A is the cross-sectional area of the flange and d is the distance from the midplane of each flange to the xy -midplane of the built-up beam. The equivalent flexure modulus is found by dividing the right-hand side of Equation 13 by I for the entire beam cross section. The stiffness of the web is given by Equation 37, where $E_{x,web}$ is the tensile modulus of the web, derived from Equation 10, t_{web} is the web thickness, and h_{web} is the web height. The flange stiffness is given by Equation 40, where κ_x is the curvature in the x -direction of a single flange under a unit applied bending moment.

$$(EI)_{total} = 2E_{m,flange}(I_{flange} + Ad^2) + (EI)_{web} \quad (13)$$

Note that

$$E_{m,flange} = \frac{1}{a_{11,flange} * t_{flange}} \quad (14)$$

$$(EI)_{web} = \frac{E_{x,web}t_{web}h_{web}^3}{12} \quad (15)$$

The strain distribution through the beam is calculated using the Timoshenko-Ehrenfest beam theory with the equivalent engineering constants calculated above. This model accounts for the shear deformation expected in bending of thick beams [7]. The beam with span L is simply supported with a symmetrical distributed transverse load $q(x)$ across a length s of the mid-span. Taking $x = 0$ to be the mid-span, the bending moment $M(x)$ in the portion of the beam under the distributed load is given by Equation 16.

$$M_x = \frac{q}{2} \left(-x^2 - \frac{s^2}{4} + \frac{Ls}{2} \right) \quad (16)$$

The strain in the x-direction as a function of distance z from the neutral axis is then determined by Equation 17. Here, E represents the equivalent flexure modulus of the beam, and I is the second moment of inertia of the beam cross-section about the neutral axis.

$$\varepsilon_x = \frac{-zM_x}{EI} = \frac{-zq(x)}{2EI} \left(-x^2 + \frac{s^2}{4} \right) \quad (17)$$

The strain in the y-direction is assumed to be negligible. The remaining component of interest is shear strain, γ_{xz} . Equation 18 describes this strain as a function of x . Variations in shear strain in the z-direction are neglected. G is the shear modulus for the web laminate, A is the beam cross-sectional area, and k is the Timoshenko shear correction factor, which is approximately 0.5 for the I-section beam.

$$\gamma_{xy} = \frac{-xq(x)}{kAG} \quad (18)$$

The strain distribution through the beam is used to compute stress in each ply of the web and flange. The global strain values are transformed into material coordinates for each ply, and each resultant stress is compared to the appropriate failure stress. First ply failure (FPF) is determined by the maximum stress criterion where the web is tested in pure shear and the flanges are tested in compression and tension. The Nomex honeycomb is placed a disadvantageous orientation where shear is applied transverse to the cells. Since its shear strength in this orientation is on the order of 500 kPa, it is extremely likely to be the first "ply" to fail. For the purposes of this analysis, FPF loads and displacements are based on CFRP laminate failure rather than honeycomb failure. Buckling of the web is a concern for a thin composite I-beam, as demonstrated by ref. [8]. Buckling failure may precede calculated FPF, thus, a simplified buckling formula (Equation 19) based on ref. [9] is employed.

$$\sigma_{x,cr} = \frac{k_b D_{11} \pi^2}{tb^2} \quad (19)$$

Peak z-displacement is estimated using Equation 20. The first component takes into

account bending deflection and the second accounts for shear deflection, as per the Timoshenko model, assuming the load can be modeled as a point load in this case. This assumption is justified since the reduction in maximum bending moment due to the load distribution is trivial.

$$w = \frac{qsL^3}{48EI} + \frac{qsL}{4kGA} \quad (20)$$

Optimal laminate stacking sequences are determined iteratively. The intent is to orient fibers along the directions of principal stress. In this case, the web is expected to take primarily shear loading, as shown in Figure 4, which corresponds to principal stresses at 45° , while the flanges experience primarily tensile and compressive stresses along the 0° direction. Thus, a symmetric $\pm 45^\circ$ pattern is chosen for the web and all 0° plies are applied to the outer faces of the flanges.

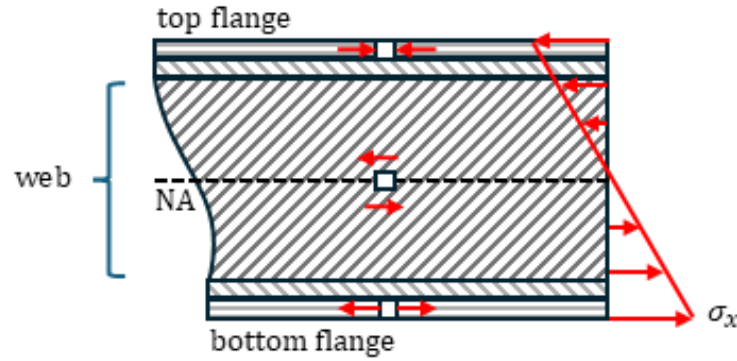


Figure 4: Diagram of the I-beam internal stress. Representative stress elements are shown in the top and bottom flanges and at the center of the web. The web is primarily in shear; the flanges are primarily in tension and compression.

1.4.1 Design Results

The first step of the analysis is to gather basic material properties for the CFRP prepreg and Nomex honeycomb core. Data provided in the official data sheets are supplemented with findings from literature and calculations. The results are summarized in Tables 1 and 2.

Table 1: Prepreg Properties

Property	Value	Source
Fiber Volume Fraction	0.55	[6], [10]
Tensile Modulus E_1 [GPa]	129.6	[6]
Tensile Modulus E_2 [GPa]	12.97	Halpin-Tsai
Shear Modulus G_{12} [GPa]	5.0	[11]
Poisson's Ratio	0.22	[12]
Ply Thickness [mm]	0.15	[6]
Density [g/m ²]	283	[6]
Tensile Failure Stress F_{t1} [GPa]	2.23	[6]
Compressive Failure Stress F_{c1} [GPa]	1.338	carpet plots for 90° uni laminate [2]
Tensile Failure Stress F_{t2} [GPa]	0.223	based on carpet plot for 90° uni laminate [2]
Shear Failure Stress F_{s6} [MPa]	95.1	[6]

Properties of the Nomex honeycomb are sourced from ref. [13]. The Poisson's ratio is based on ref. [5].

Table 2: Nomex Core Properties

Property	Value
Tensile Modulus $E_1 = E_2$ [MPa]	26.7
Shear Modulus ("W" direction min) [MPa]	0.028
Shear Strength ("W" direction typical) [MPa]	0.46
Poisson's Ratio	0.24
Density [g/m ²]	512.1

Overall dimensions of the beam are specified as per manufacturing and structural requirements. While a beam height of 101.6 mm (4 in) is desired to maximize the flexural rigidity within the competition rules, the web height is restricted by available tooling to 76.2 mm (3 in). The thickness of the web core is 11 mm (0.42 in), which is twice the thickness of a single layer of Nomex honeycomb sheet. Laminate stacking sequences are specified as $[0]_6$ for the flange reinforcement and $[\pm 45]_{3s}$ for the web facings. The resulting built-up flange thickness is 1.8 mm (0.07 in) and the web facings are 0.9 mm (0.04 in). This results in an overall beam height of 81.6 mm (3.21 in) and web thickness of 12.5 mm (0.5 in).

Using the dimensions and areal weights of prepreg and honeycomb core, the total weight of the bridge is predicted to be 348 g. This is very light, considering past SAMPE competitions had imposed a 600 g weight limit to the CFRP I-beam category [8].

As described in the design analysis section, the strains in the beam are determined using equivalent elastic constants for each component. The values of these constants are summarized in Table 3

Table 3: Estimated Beam Properties

Property	Value
Tensile Modulus (web) $E_{x,web}$ [GPa]	17.03
Shear Modulus (web) G_{xz} [GPa]	26.43
Tensile Modulus (flange) $E_{x,flange}$ [GPa]	24.87
Transverse Tensile Modulus (flange) $E_{y,flange}$ [GPa]	11.75
Shear Modulus (flange) G_{xy} [GPa]	8.55
Flexure Modulus (beam) E_{flex} [GPa]	45.95

Note that E_y for the web is taken to be approximately equal to E_x since it contains a balance of 45° and -45° plies.

Predicted displacement at FPF for 3-point bending is 4.45 mm under an applied load of 7700 lb (31.1 kN). The first ply failure results for 7700 lb of applied load are summarized in Tables 4 – 6. The failure mode based on the maximum stress criterion is compression in the flange, both in the axial and transverse directions. Buckling failure in the web is not predicted to arise prior to FPF in the flanges. FPF in the web does not occur until the applied load exceeds 100 kips (444 kN). Shear failure of the Nomex honeycomb is challenging to model in detail. Using "W" direction shear strength alone, the Nomex is predicted to fail under an applied load as low as 1000 lb (4.4 kN), well before reaching the beam design load. Based on this analysis, it is predicted that compressive failure in the upper flange will be the first failure mode to occur, not including shear failure of the Nomex honeycomb. Note that in the original report, the flexural modulus was over predicted, resulting in an extremely high FPF on the order of 40 kips. With the corrected flexural modulus, FPF occurs just after the target load is reached. Thus, it will be important for any future beam assemblies to have additional reinforcement, particularly in the upper flange.

Table 4: Top Flange Compressive FPF. $R_i = \sigma_i/\sigma_{if}$ is the ratio of stress in the ply to the failure stress in the direction i ($i=x, y$, or s where s stands for shear).

Ply		Rx	Ry	Rs
1	Top	-1.000	-0.220	0.0
2		-0.998	-0.220	0.0
3		-0.996	-0.219	0.0
4		-0.994	-0.219	0.0
5		-0.992	-0.218	0.0
6		-0.990	-0.218	0.0
7		-0.504	-0.603	-0.624
8		-0.504	-0.602	0.622
9		-0.503	-0.600	-0.621
10		-0.502	-0.599	0.620
11		-0.501	-0.598	-0.618
12	Bottom	-0.500	-0.597	0.617

Table 5: Web FPF

Ply		Rx	Ry	Rs
1	Right	0.033	-0.027	0.0
2		-0.023	0.019	0.0
3		0.033	-0.027	0.0
4		-0.023	0.019	0.0
5		0.033	-0.027	0.0
6		-0.023	0.019	0.0
7	Core	0.0	0.0	-0.043
8		-0.023	0.019	0.0
9		0.033	-0.027	0.0
10		-0.023	0.019	0.0
11		0.033	-0.027	0.0
12		-0.023	0.019	0.0
13	Left	0.033	-0.027	0.0

Table 6: Flange Tensile FPF

Ply		Rx	Ry	Rs
1	Top	0.350	0.418	-0.617
2		0.350	0.418	0.618
3		0.351	0.419	-0.619
4		0.352	0.420	0.621
5		0.353	0.421	-0.622
6		0.353	0.422	0.624
7		0.693	0.153	0.0
8		0.694	0.153	0.0
9		0.696	0.153	0.0
10		0.697	0.153	0.0
11		0.699	0.154	0.0
12	Bottom	0.700	0.154	0.0

1.5 Design Risk and Sensitivity

1.5.1 Design Parameters vs Performance Sensitivity

1. Total Bending Stiffness (EI) Sensitivity

The total bending stiffness of the beam is its resistance to bending and is given by

$$EI = E_{\text{flex}} I_{\text{total}}, \quad (21)$$

where E_{flex} is the effective flexural modulus of the composite section and I_{total} is the second moment of area of the built-up I-beam cross section. The total second moment

of area is computed as

$$I_{\text{total}} = I_{\text{web}} + 2(I_f + A_f d^2), \quad (22)$$

where

$$I_f = \frac{1}{12} w t_f^3, \quad (23)$$

$$A_f = w t_f, \quad (24)$$

$$I_{\text{web}} = \frac{1}{12} t_w (h - 2t_f)^3, \quad (25)$$

$$d = \frac{h}{2} - \frac{t_f}{2}. \quad (26)$$

$$(27)$$

Sensitivity was evaluated by perturbing each geometric parameter by 2% and computing the resulting percent change in total bending stiffness:

$$\text{Sensitivity} = \frac{EI(x(1 + \delta)) - EI(x)}{EI(x)} \times 100\%. \quad (28)$$

Using the computed values for a 2% increase in beam height, an example calculation for obtaining sensitivity is shown below:

$$EI_{\text{base}} = 339784.3090400041 \quad (29)$$

$$EI_h = 354794.69795160706 \quad (30)$$

the percent change in bending stiffness is

$$\text{Sensitivity}_h = \frac{EI_h - EI_{\text{base}}}{EI_{\text{base}}} \times 100\% \quad (31)$$

$$= \frac{354794.69795160706 - 339784.3090400041}{339784.3090400041} \times 100\% \quad (32)$$

$$= \frac{15010.38891160296}{339784.3090400041} \times 100\% \quad (33)$$

$$= 4.41762274\%. \quad (34)$$

The results indicate that total bending stiffness is most sensitive to beam height h , as a 2% increase in h resulted in a 4.418% increase in EI . In comparison, flange thickness t_f produced a -3.647% change, web thickness t_w produced a 0.203% change, and beam width w produced no measurable change in EI under the current formulation. The negligible sensitivity to width arises from cancellation between geometric scaling in I_{total} and the corresponding scaling within the effective modulus calculation.

Overall, beam height is the dominant geometric parameter governing bending stiffness due to its strong influence on the cubic dependence of the second moment of area.

2. Total Shear Stiffness Sensitivity

For Timoshenko beam theory, the total shear stiffness is given by kAG , where k is the shear correction factor, G is the effective transverse shear modulus, and A is the effective shear area. In this case, $G = G_{xz}^{eq}$, as shear deformation occurs in the xz -plane, and the appropriate shear area is the web area, A_{web} , since the web carries the majority of the transverse shear force.

A 2% perturbation was applied to the same four geometric parameters (h , w , t_f , and t_w). The results indicate that total shear stiffness is most sensitive to beam height h , which produced a 3.032% change. In comparison, beam width w , flange thickness t_f , and web thickness t_w resulted in changes of 0.890%, 0.533%, and 0.390%, respectively. It can be observed that beam height h has the greatest influence on overall beam performance, as both total shear stiffness and total bending stiffness were most sensitive to variations in this parameter.

1.5.2 Composite Laminate Theory Violations

1. Kirchhoff–Love Assumption

CLT assumes transverse shear strains are zero, meaning normals to the mid-surface remain straight and perpendicular after deformation. This assumption is valid for thin plates where bending deformation dominates and shear-induced deflection is negligible.

In the present I-beam design, this assumption is likely violated. The web height is large relative to its thickness and carries significant transverse shear force. As a result, shear deformation in the web is not negligible. Neglecting transverse shear leads to an overestimation of bending stiffness and an underestimation of beam deflection thus affecting the performance of the beam.

2. Plane Stress Assumption

CLT assumes a plane stress condition in which only the in-plane stress components σ_{xx} , σ_{yy} , and τ_{xy} are nonzero, while the out-of-plane stresses σ_{zz} , τ_{xz} , and τ_{yz} are taken as zero. This assumption is generally valid for thin laminates subjected primarily to in-plane loading.

For the present I-beam design, this assumption is reasonable in the flat regions of the top and bottom flanges, which behave similarly to thin plates. However, the assumption breaks down near the flange–web intersection, where three-dimensional stress states develop due to load transfer between perpendicular laminates. In these regions, out-of-plane and interlaminar stresses are present, and CLT cannot capture potential failure mechanisms such as delamination, debonding, or matrix cracking which all hinder beam performance.

3. Perfect Bonding Assumption

CLT assumes perfect bonding between all plies, such that no slip or separation occurs and strain is continuous through the thickness. Under this assumption, the laminate behaves as a single, fully integrated structure.

In the present design, two C-channels are adhesively bonded together to form the I-beam. The bonded interface between the two sections may not behave as a perfectly rigid, monolithic laminate, particularly if manufacturing imperfections or adhesive compliance are present. This violates the perfect bonding assumption inherent to CLT.

As a result, load transfer across the joint may be less effective than predicted, potentially reducing the effective bending stiffness of the beam. Additionally, slip or debonding at the interface may occur prior to laminate-level failure, leading to an overestimation of structural strength if analyzed solely using CLT.

1.6 Manufacturing Inaccuracies vs Performance

Ply orientation is identified as the manufacturing parameter most susceptible to variation, as small placement errors may occur during manual layup or handling. A realistic tolerance of $\pm 2^\circ$ is assumed for the $\pm 45^\circ$ plies, such that the actual fiber orientations may range from 43° to 47° during manufacturing.

To evaluate the impact of this variation, the web laminate stacking sequence was perturbed by $+2^\circ$, and the resulting total bending stiffness EI and total shear stiffness were recalculated. The analysis indicates that this misalignment produces a -0.179% change in total bending stiffness, corresponding to a negligible reduction in flexural performance. In contrast, the same misalignment produces a -1.582% change in total shear stiffness, reflecting the dominant role of the web and its $\pm 45^\circ$ plies in carrying shear loads.

These results suggest that the beam is relatively insensitive to small ply angle deviations with respect to bending stiffness, but is more sensitive in terms of shear stiffness due to the web-dominated shear response.

1.7 Manufacturing Method Selection

Several manufacturing options are possible for layup and cure of the laminate I-beam. The first option is low-pressure oven cure, which is inexpensive and provides the necessary temperatures for resin cure. However, since the oven does not apply pressure to the mold, the plies are not consolidated and the resulting part is weaker than if a hot press or autoclave were used. A second option is to use a hot press. The hot press can apply pressure and heat to the flanges of the beam, but the web would not cure properly since it would be reliant on heat transfer from the flanges. This is very inefficient and may not be effective at all due to the mold acting as a heat sink and pulling heat out of the web before it could even reach the middle of the web. A specialized

mold could be designed apply heat and pressure along multiple axes during cure in the hot press, but the tooling for this would be exceedingly complex for this project. A third alternative is resin infusion, which would improve production speed since all of the plies could be set without the resin, allowing for adjustments. Since this project is restricted to using prepreg, a gradual layup process is required to avoid mistakes since changes cannot be made after the plies have been set in place. The selected processing method is autoclave cure of prepreg in a custom mold. This method provides uniform heat and pressure across the mold geometry without the need for expensive tooling.

To build the mold, two avenues were examined: machining molds from full bars of metal and adapting existing C-channels of metal to our needs. Although machining the molds from solid bars enables more geometric flexibility, the material and fabrication costs would be quite high. Instead, a prefabricated C-channel of mild steel is found on McMaster-Carr. A cross section of the channel is shown in figure 5. The drawback of using this cross section is the lower corner radii of only 0.08 in (2.26 mm). These would produce high stress concentrations at the web-flange interfaces. To reduce the stress concentration in the corner, the radius must be increased. Per section 3.2 of the 2025 SAMPE bridge contest rules, the maximum radius between the web and flange is 0.5 in (12.7 mm) [1]. The closest available endmill sizes are 5/16 in (11.11 mm) and 3/8 in (9.53 mm) radius. A 3/8 in mill is selected since the chuck required for this size is available in the machine shop and the fillet size is large enough to reduce the stress concentration at the corner without risking disqualification for being oversized. Figure 6 shows the cross section of the channel with the modified corner radius.

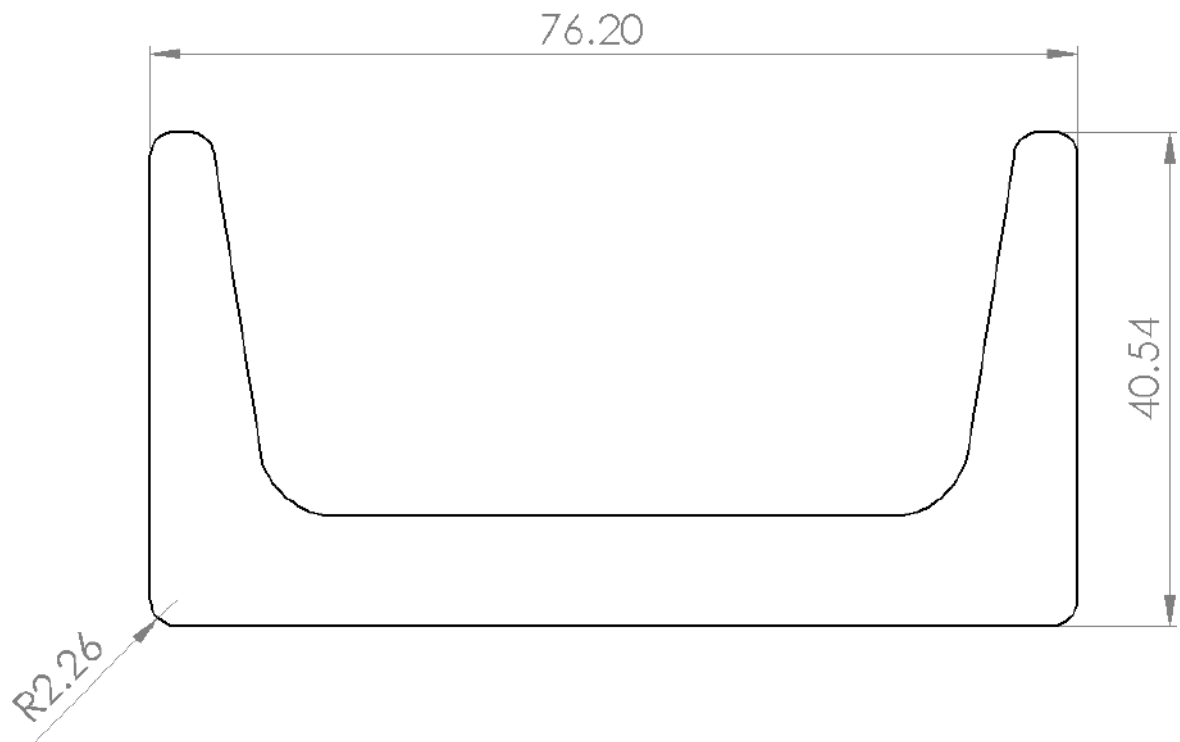


Figure 5: Original

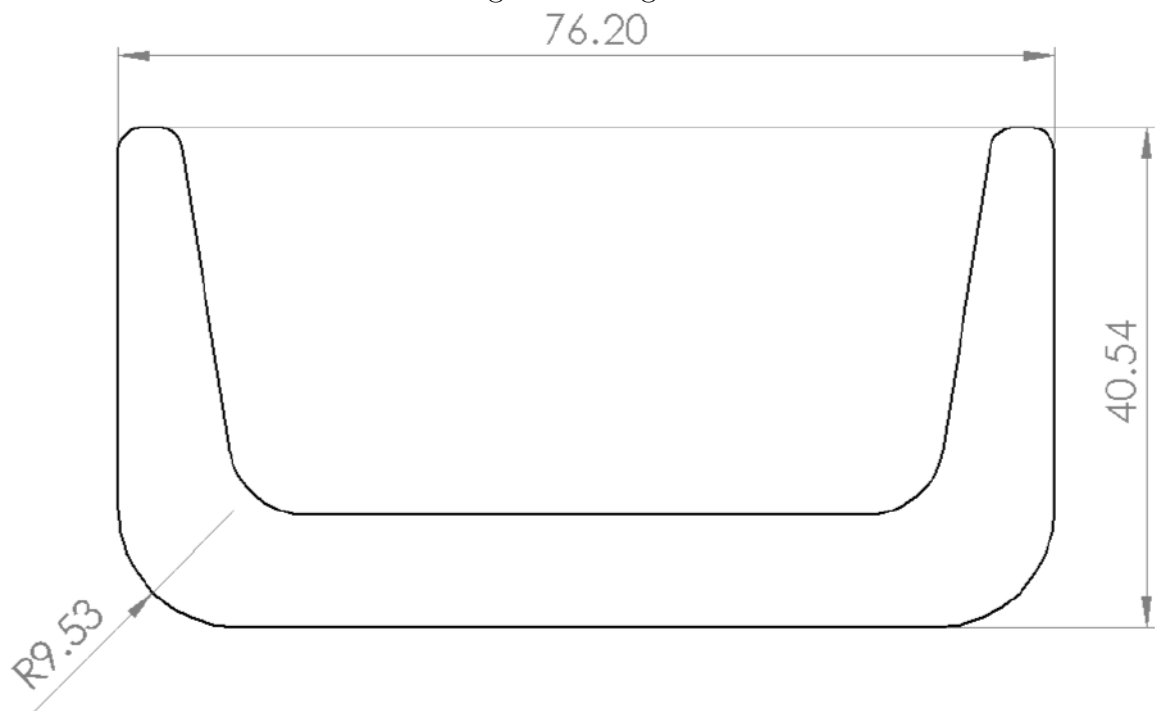


Figure 6: Mold cross section with increased corner radius

Figure 7: Combined figure

The second modification required for the channels is a pair of holes to help align each side during assembly. As the team noticed during the warpage lab, once the prepreg

is mounted to the mold, it will be challenging to remove for re-alignment. Thus, to ensure the molds come together as expected, a dowel pin will be inserted into a hole in the channel on each side of the layup.

The final component to consider is the core. Since the web of the beam is susceptible to buckling, adding additional material to the middle of the web can help to stabilize the outer composite layers. This is an important role, but the core does not need to carry incredibly high loads, the composite should handle that. Instead, the core material should be as lightweight as possible, so aluminum or aramid honeycombs are commonly used for this purpose much like in a conventional sandwich panel. Since weight is an important parameter for the beam, the Nomex honeycomb was selected over the aluminum. To further improve the strength of the core, we are filling the honeycomb with expanding polyurethane foam. This will help to reinforce the honeycomb and prevent it from collapsing internally. Since the Nomex honeycomb is only 5.33 mm (0.21 in) thick and we are allowed a web up to 15.24 mm (0.6 in) thick, we will be using two layers of honeycomb which will be placed on top of one another. They will be bonded together using the expanding foam and after the foam has cured, any excess will be machined off the top surface and the edges will be cut to dimension.

Two primary approaches could be employed to improve the performance of the beam. The first involves the use of finite element analysis to optimize the beam geometry. The competition rules permit varying cross-sections for the CFRP I-beam category, which opens the door to building arched or bowed bridges. However, to simplify manufacturing and design analysis, a prismatic beam was preferred. With FEA software, more complex geometry could be evaluated, and material distribution could be optimized.

The second opportunity for improvement lies in the selection of core materials. Foam-filled honeycomb does not represent the most weight-efficient solution. Alternative materials, such as aerogels, could provide comparable structural support at a lower weight. Such high-tech materials are costly, and applying them to the project would provide marginal benefit aside from clout.

1.8 Manufacturing Plan

The manufacturing plan is structured to ensure that the laminate stacking sequences selected in the design phase are accurately implemented in the fabricated beam. Because the structural response of the I-beam depends strongly on the directional stiffness of the composite laminates, careful control of ply orientation, stacking symmetry, and thickness buildup is required. The web and flange laminates serve distinct structural roles, and their layups are therefore treated separately during fabrication.

1.8.1 Web Laminates

The web is constructed as a balanced symmetric angle-ply laminate, including two core layers. Rather than creating four flat laminates and joining them into an I-shape,

the web facings are each wrapped around the C-channel molds to ensure a smooth transition to the flanges. The stacking sequence is $[\pm 45]_{3s}$ for each face for a total of six plies on each side of the Nomex honeycomb core. The $\pm 45^\circ$ orientation aligns fibers with the principal shear stress directions expected in the web under transverse loading. Symmetry about the laminate mid-plane (including the Nomex core) eliminates the B coupling matrix terms predicted by CLT, thereby preventing unintended extensional–bending coupling effects and warpage.

During layup, fiber angles are verified prior to placement to maintain an orientation tolerance of $\pm 2^\circ$, as deviations beyond this range were shown to reduce shear stiffness in the sensitivity analysis. Particular attention is given to ply conformity along web-flange transition region, as fiber wrinkling in this region would locally alter the effective shear modulus. The final cured web laminate thickness corresponds to the designed 0.9 mm facing thickness on each side of the honeycomb core.

1.8.2 Flange Laminates

The flange laminates are fabricated using unidirectional $[0]_6$ stacking sequence. These plies are applied on the top and bottom of the beam after the web is laid up, and they connect to the rolled-over angle plies. Thus the overall flange laminates are asymmetric, and warpage is possible. The 0° plies are continuous and oriented in the principal stress direction for the top and bottom of the beam in bending. The beam layup is illustrated in Figure 8.

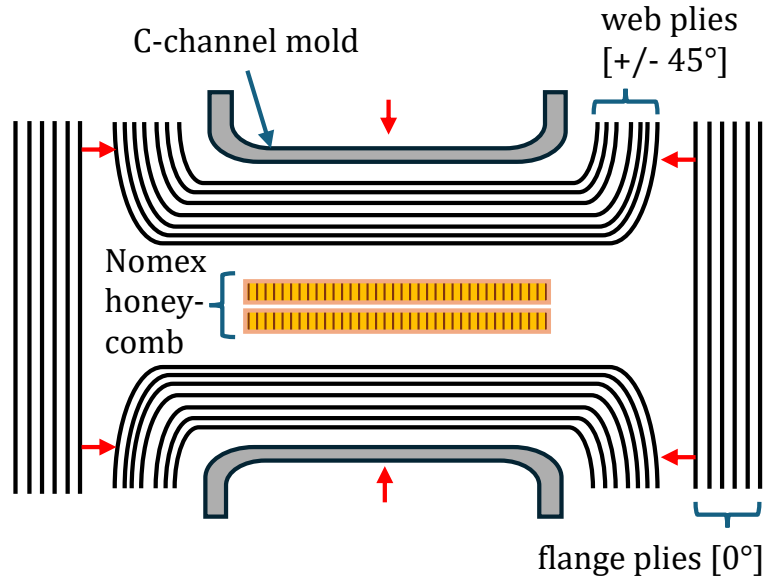


Figure 8: Layup configuration for the CFRP prepreg and Nomex I-beam

1.8.3 Assembly and Layup Procedure

1. Core Preparation and Placement

Two honeycomb sheets are bonded together using expanding polyurethane foam to achieve the required web thickness. After curing, excess foam is machined flush and the core is trimmed precisely to final dimensions. A layer of Henkel 9696 adhesive film is placed on top of the lower web laminate. The prepared honeycomb core is then positioned carefully to ensure uniform surface contact, centered placement within the channel, and no localized crushing of the honeycomb cells. A second adhesive film layer is applied to the top surface of the core to prepare for bonding with the upper web laminate.

2. Tool Preparation

Both modified steel C-channel molds are cleaned thoroughly using solvent to remove contaminants that could interfere with consolidation or surface finish. A high-temperature mold release agent is applied uniformly to all interior surfaces. Release film is carefully placed inside the mold cavity, ensuring proper conformity along the 9.53 mm corner radii to prevent bridging or wrinkling. The dowel pin holes in both channels are inspected to confirm dimensional accuracy. Because prepreg tack limits repositioning once contact is made, precise alignment during closure is critical to maintaining geometric accuracy and fiber orientation.

3. Layup of the Web Laminate

The web laminate is laid up first inside the lower C-channel. The layup sequence proceeds as follows:

- Place the first $+45^\circ$ ply into the mold.
- Consolidate lightly with rollers/squeegee to remove entrapped air.
- Place the second ply at -45° , carefully maintaining alignment.
- Continue alternating $+45^\circ$ and -45° plies until six plies are stacked.
- Repeat the alternating sequence to complete the symmetric half of the laminate.
- The same procedure is repeated for the upper C-channel.

This concludes the layup sequence for the web laminate.

5. Channel Closure and Alignment

Once both web laminates and the core are prepared, the upper channel is inverted and aligned using the precision dowel pins. This step is performed slowly to prevent disturbance of the prepreg or adhesive layers. The upper channel is lowered gradually until contact is made with the adhesive film on the core. Light manual pressure is applied to promote uniform contact across the bonding surfaces. At this stage, the web sandwich structure is fully assembled but not yet consolidated under cure pressure.

6. Layup of the Flange Reinforcement

After the channels are joined, flange reinforcement is applied externally. The flange stacking sequence is $[0]_6$, hence the layup proceeds as follows:

- Place the first 0° ply across the top flange surface.
- Ensure uniform contact with the underlying web laminate.
- Consolidate lightly with rollers/squeegee to remove entrapped air.
- Continue sequential placement of the remaining five 0° plies.
- The same procedure is repeated for the bottom flange.

This concludes the layup sequence for the flanges.

1.8.4 Curing Procedure

At this stage, the complete laminate architecture consists of symmetric $\pm 45^\circ$ web laminates optimized for shear, honeycomb/foam sandwich core for weight saving, and unidirectional 0° flange reinforcement optimized for bending. The curing stage is critical to ensuring that the fabricated laminate achieves the mechanical properties assumed in the structural analysis. To cure the beam, the entire beam and mold assembly will be placed in a vacuum bag, sealed, and placed under a vacuum. Once a proper seal has been verified, it will be placed in the autoclave at 125°C (257°F) and under 30 psi of pressure for curing.

Calculating the cure time requires analysis of the steel mold. The steel introduces a thermal resistance between the autoclave chamber temperature and the prepreg temperature. The prepreg datasheet specifies a cure time of one hour at 250°F , but modification will be required to account for the rate of heat flux to the innermost plies.

The time required for the innermost composite ply to reach 120°C is approximated using one-dimensional transient heat conduction analysis through a multi-layer slab consisting of 9 mm of steel mold and 2 mm of composite laminate. The steel layer is treated as a lumped thermal mass, while the composite layer is modeled using the transient plane wall solution characterized by the Fourier and Biot numbers. The internal temperature rise is estimated using the Heisler chart solution for a finite slab subjected to convective boundary conditions at 125°C .

1.8.5 Governing Equation

The objective is to determine the time required for the innermost composite ply to reach $T(L,t) = 120^\circ\text{C}$ when the autoclave ambient temperature is maintained at $T_\infty = 125^\circ\text{C}$. The initial temperature of the entire system is assumed uniform at $T_i = 25^\circ\text{C}$. Transient one-dimensional heat conduction in each layer is governed by

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

where thermal diffusivity is defined as

$$\alpha = \frac{k}{\rho c_p}$$

1.8.6 Material Properties

The following values are material thermal properties for the steel mold.

$$k_s = 50 \text{ W/m}\cdot\text{K}$$

$$\rho_s = 7850 \text{ kg/m}^3$$

$$c_{p,s} = 470 \text{ J/kg}\cdot\text{K}$$

$$\alpha_s = \frac{50}{7850 \times 470} = 1.36 \times 10^{-5} \text{ m}^2/\text{s}$$

While the following values are material thermal properties for the composite prepreg.

$$k_c = 0.2 \text{ W/m}\cdot\text{K}$$

$$\rho_c = 1600 \text{ kg/m}^3$$

$$c_{p,c} = 1000 \text{ J/kg}\cdot\text{K}$$

$$\alpha_c = \frac{0.2}{1600 \times 1000} = 1.25 \times 10^{-7} \text{ m}^2/\text{s}$$

1.8.7 Steel Mold Heating Time

Using the lumped capacitance solution,

$$\frac{T - T_\infty}{T_i - T_\infty} = e^{-t/\tau}$$

where the time constant is

$$\tau = \frac{\rho c_p L}{h}$$

$$\tau_s = \frac{7850 \times 470 \times 0.009}{50}$$

$$\tau_s = 664 \text{ s}$$

The steel reaches approximately 95% of the ambient temperature at

$$t \approx 3\tau_s$$

$$t_s \approx 2000 \text{ s}$$

$$t_s \approx 33 \text{ minutes}$$

1.8.8 Composite Prepreg Heating Time

Once the steel approaches 125°C, the composite surface may be treated as suddenly exposed to $T_s = 125^\circ\text{C}$. Assuming an autoclave convection coefficient of

$$h = 50 \text{ W/m}^2\text{K},$$

the Biot number is

$$Bi_c = \frac{50 \times 0.002}{0.2} = 0.5$$

The required dimensionless temperature at the back surface for $T = 125^\circ\text{C}$ is

$$\theta = \frac{120 - 125}{25 - 125} = 0.05$$

For the prepreg back surface with $Bi_c = 0.5$ and $\theta = 0.05$, the Heisler chart gives approximately

$$Fo \approx 1.8$$

The Fourier number is defined as

$$Fo = \frac{\alpha t}{L^2}$$

Thus, we can solve for time

$$t \approx \frac{FoL^2}{\alpha}$$

$$t \approx \frac{1.8(0.002)^2}{1.25 \times 10^{-7}} \approx 57.6 \text{ seconds}$$

1.8.9 Total Heating Time

The steel mold reaches curing temperature in approximately 33 to 34 minutes while the composite reaches temperature in approximately 1 minute once the steel is hot. Therefore, the total time required for the innermost composite ply to reach 120°C is approximately

$$t_{\text{total}} \approx 35 \text{ minutes}$$

after the autoclave ambient reaches 125°C. Once the innermost composite ply reaches the required cure temperature, the laminate must be held at this temperature for one hour, in accordance with the prepreg manufacturer's datasheet specifications. Following that, the structure must undergo a cooling cycle within the autoclave. The cooling rate should be gradual to minimize thermal gradients between the steel mold and the composite laminate, thereby reducing the risk of residual stresses, microcracking, or distortion. The laminate should not be retrieved until it has cooled sufficiently and reached a safe handling temperature, ensuring the matrix has fully solidified.

1.8.10 Post Processing

Once the beam has been removed from the molds, it may require some minor post processing. This is all minor, such as removing any epoxy flash from the edges of the beam, making sure it still meets all of the geometric requirements described in the SAMPE rule set, and trimming the ends to the correct length. However, there should not be any significant work required.

1.9 Conclusions and Concerns

This section details the design of a carbon fiber reinforced polymer I-beam for the SAMPE 2026 composite bridge competition. The beam is intended to withstand a load of 7200 lbf in static 3-point bending. Using classical laminate theory and Timoshenko-Ehrenfest beam theory, a theoretical beam model is produced to estimate

the deformation and first-ply-failure of the beam. The sensitivity of the structure to variations in layup and material properties are considered as well as alternative manufacturing methods.

The primary concerns are premature failure due to manufacturing defects and unanticipated behavior of the sandwich core. While the analysis predicts a very high first-ply-failure load of 45000 lb, many variables are present in the fabrication process, and careful design and assembly is crucial to ensure the beam meets the structural requirements. For example, if the resin does not completely cure due to incorrectly calculated heat transfer through the steel mold, the resulting part may be weaker than expected. Additionally, if the layups are not smooth, the small wrinkles may result in amplified internal stresses that cause the beam to crack before reaching the target load.

2 Manufacturing Results

2.1 Initial Builds

Early attempts to build the beam according to the original manufacturing plan revealed areas of improvement and important changes required. The first challenge was the foam fill for the Nomex honeycomb core. The core was constructed from 0.21 in (5.3 mm) meta-aramid (Nomex) honeycomb cut into two 3 x 24 in (76.2 x 609.6 mm) rectangular sheets. The cut sheets were stacked and filled with polyurethane expanding foam. As shown in Fig. 9, the curing foam formed a rough surface on the beam. As the foam expanded, it forced the honeycomb sheets apart as shown in Fig. 10. Although the addition of the foam resulted in a stiffer core, the uneven surface was not easily leveled, and the final structure was not sufficiently flat to interface with the laminate facings.

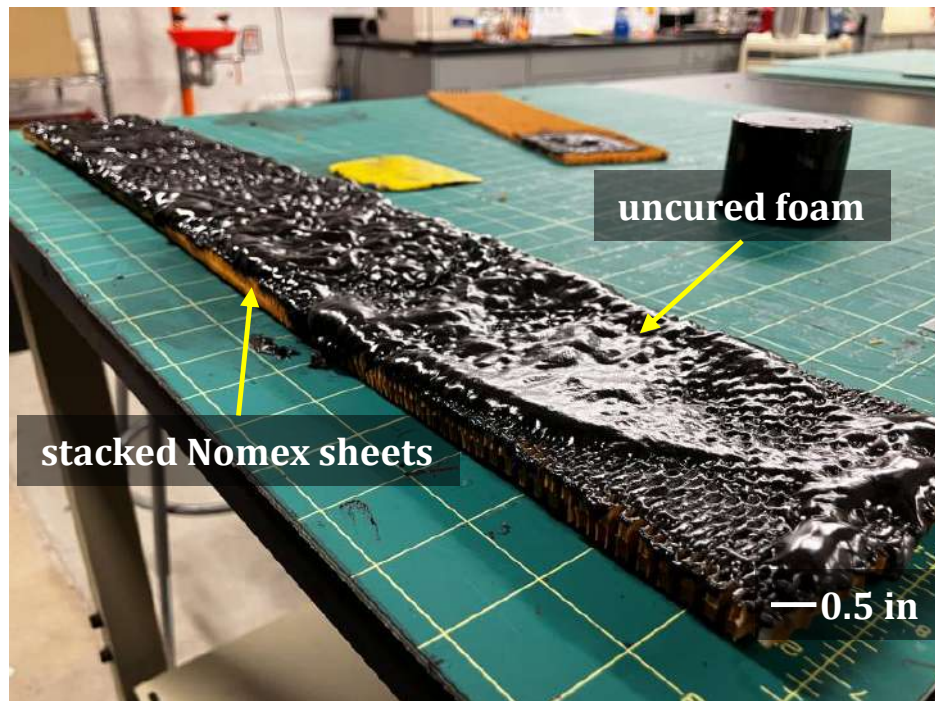


Figure 9: Uncured polyurethane foam filled Nomex honeycomb panels



Figure 10: Side view of cured polyurethane foam filled Nomex honeycomb with foam expanded between the layers. The top of the piece is sanded but retains the curvature introduced by the sandwiched foam.

For the final beam assembly, the foam was eliminated to provide a level surface for the laminate bonds. Violet epoxy film adhesive (Hysol EA 9696 .06) was applied between the Nomex layers. This adhesive was also used to attach the honeycomb to the laminates on either side.

The second challenge was related to the bagging and cure process. The first iteration of the beam was laid up and cured according to the manufacturing plan (section 1.8.3) excluding the foam fill and mold release agent. The vacuum bag was adequate to release the cured composite from the mold.

To keep the mold C-channels securely connected to each other via gravity, the beam was cured on its side with neither flange in contact with the platform in the autoclave. The autoclave pressure was set to 5 psi, and the temperature was held at 125°C for 1 hour and 45 minutes. Autoclave pressure was expected to be sufficient to keep the prepreg pressed against the mold. However, during the curing process, the unrestrained flanges relaxed and were pulled outward, becoming nearly parallel to the core faces, as shown in Figure 11.

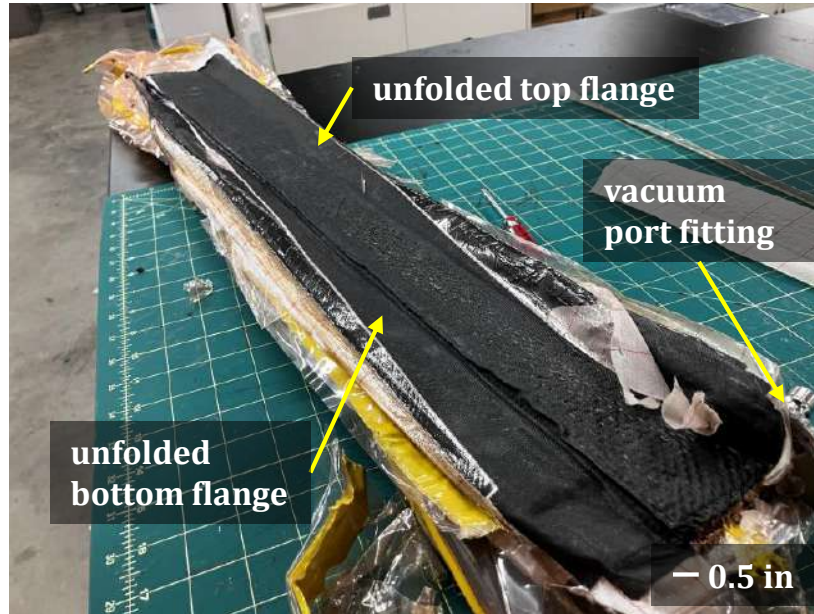


Figure 11: First attempt at curing the I-beam. Significant warpage is observed in the flanges.

The extreme flange warpage was likely the result of unconstrained flange laminates softening in the two-piece vacuum bag. The bag construction was biased toward flattening the entire beam into the plane of the web core panels, as it was placed between the C-channels and connected above the top flange and below the bottom flange. As pressure was applied, the bag may have initially conformed to the molded flange, but when the viscosity of the epoxy resin in the prepreg dropped with the increasing temperature, the bag was free to flatten with the unconstrained flanges inside.

To resolve the issue, a second iteration of the beam was built using a modified mold and cure process. To accelerate the assembly, each 6-ply laminate was constructed separately and then placed around the mold. The angle-ply cut for the web were cut to achieve a staggered pattern, as the available width of prepreg was not sufficient to cover the full beam length. This construction is illustrated in Figure 12. The team also considered using a stackup of alternating $\pm 45^\circ$ plies in the layer 1 configuration shown in Figure 12, simply to preserve material continuity in the center of the beam where load is applied. However, the final approach is preferred, as each alternating layer can support the adjacent layers where prepreg interfaces are present.

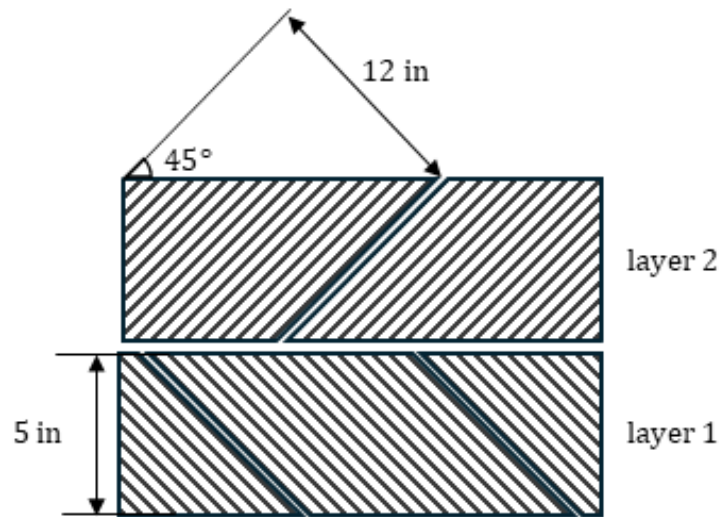


Figure 12: Cutting scheme for the web laminates. Layer 1 and layer 2 are stacked in an alternating pattern three times per web facing.

To prevent unwanted distortion, particularly in the flanges, the autoclave pressure was reduced to about 0 psi, and the temperature was reduced to 115°C. The beam was cured upright with the lower flange constrained by the autoclave platform and the upper flange constrained by an aluminum plate. The two mold C-channels were joined using C-clamps.

The first build therefore served as a process-development trial rather than a final specimen. The two most significant lessons were that the expanding foam created an unsuitable bonding surface for the web laminates and that the flange geometry required stronger mechanical constraint during cure. Based on these findings, the second build was modified by removing the foam-filled concept, using adhesive film for bonding, and physically constraining the flanges during cure. These changes resulted in a more stable and manufacturable beam that could be tested successfully.

The assembly process for the final beam is summarized in Figures 13 and 14.

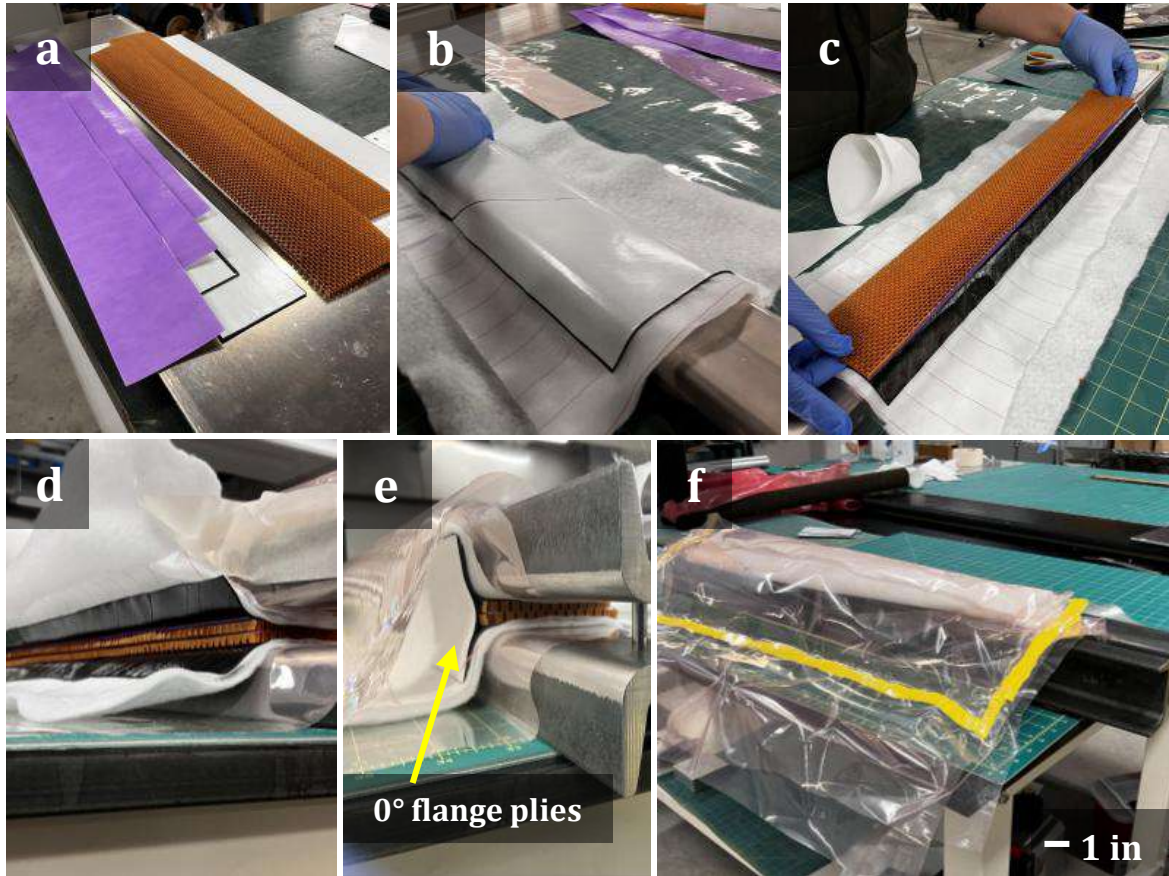


Figure 13: Summary of assembly steps. (a): Lamina are cut and stacked for the web and flange. Adhesive and Nomex honeycomb are cut to size for the core. (b): The mold is covered by the vacuum bagging film, bleeder cloth, and peel ply. The web laminates are placed on top of the bleeder cloth and wrapped around the mold. (c): Adhesive is mounted on top of the web laminate, and the Nomex is connected to the adhesive. (d): The process is repeated for the other half. Adhesive is placed on top of the first Nomex layer. The second assembly is inverted on top of the first with the adhesive between them. (e): Unidirectional flange laminates are attached to each side of the assembly. (f): The beam is rotated upright, and the bags are joined with vacuum bagging tape.

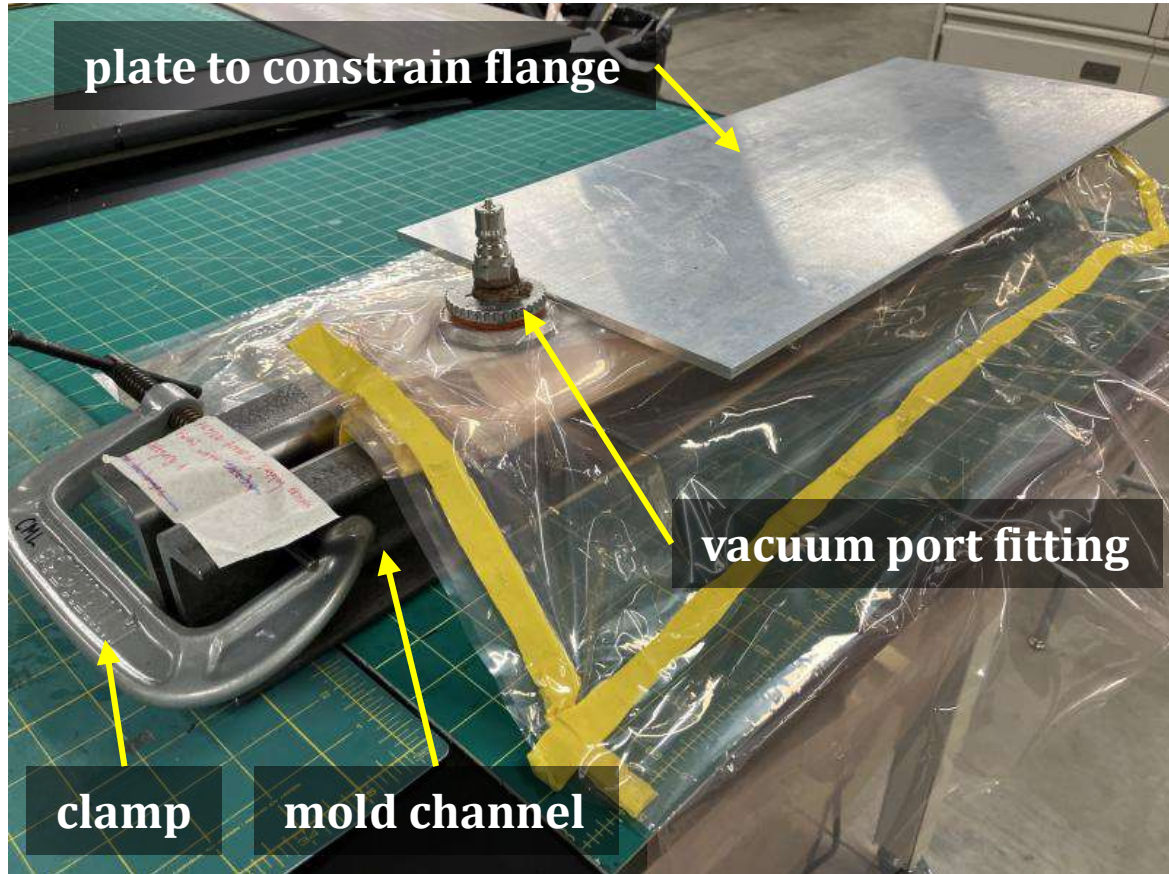


Figure 14: Bagged and clamped beam assembly prepared for curing in the autoclave.

2.2 Physical Properties

The final manufactured beam weighed 432.4 g and measured $24 \times 3 \times 2.8$ in. The average flange thickness was 0.08 in, while the web thickness, including the core, was 0.48 in. Overall, the beam was successfully fabricated as a testable specimen, but some geometric imperfections remained from the curing and bagging process.

Flange rippling was observed along the beam length, as shown Fig. 15. This effect was most pronounced on the vacuum-port side, where the adapter plate and offset top plate interfered with uniform consolidation. In contrast, the flange surface against the autoclave platform remained flatter because it was fully constrained by the platform and mold channel. The web geometry was more uniform than the flanges because the clamped mold channels and honeycomb-supported web better resisted local deformation during cure.

These observations indicate that the final beam achieved the intended overall architecture, but the flange surfaces were more sensitive than the web to tooling interference, bagging details, and local consolidation conditions. Future builds would benefit from relocating the vacuum connector away from the beam and extending breather cloth coverage to improve pressure distribution across the flange surfaces.



Figure 15: Ripples and inconsistent geometry in the flanges are noticeable when looking down the length of the beam. The web however has a uniform thickness due to the honeycomb support and the moulds.

3 Testing Results

3.1 Objectives

The beam was tested in three-point bending using a setup representative of the SAMPE bridge loading configuration. The method is related to ASTM D790-17 in that a simply supported specimen is loaded transversely at midspan while load and displacement are recorded to characterize flexural response [14]. However, the present specimen is a built-up composite I-beam rather than a standard rectangular coupon, so the analysis was adapted to account for the measured beam geometry and the separate stiffness contributions of the web and flanges.

The objectives of the test were to determine the equivalent flexural modulus of the beam (E_x), estimate the flexural strength at the first major failure event (σ_{fM}), and measure the maximum deflection prior to failure (δ_f). Load and displacement data from the initial linear portion of the response were used to determine the effective bending stiffness of the beam. These results were then combined with measured cross-sectional dimensions and transformed-section calculations to estimate the equivalent beam modulus and the stresses in the flange and web at the onset of failure.

3.2 Equipment

To perform the destructive test, the 125 kN (22 kip) 4-pole MTS testing frame at the Newmark Structural Engineering Laboratory (NSEL) at the University of Illinois, Urbana-Champaign was used. The loading span was 23 in and the crosshead speed was approximately 1 in/min. The bridge was supported by round bars at each end, as shown in Fig. 16. All testing was carried out on March 17, 2026.



Figure 16: I beam testing setup at NSEL on the 125 kN test frame

3.3 Assumptions

As shown in Fig. 17, there are two load peaks due to progressive failure following initial breakage. For mechanical property determination, only the initial failure point is used. Toe region compensation is carried out assuming the toe of the load-displacement curve is purely a consequence the beam seating and is not relevant to the composite material properties (see Fig. 17). For stress and strain calculations, the beam is assumed to behave as a simply supported Euler beam with different effective elastic moduli for the web and flanges. The strain distribution through the height is assumed to be linear.

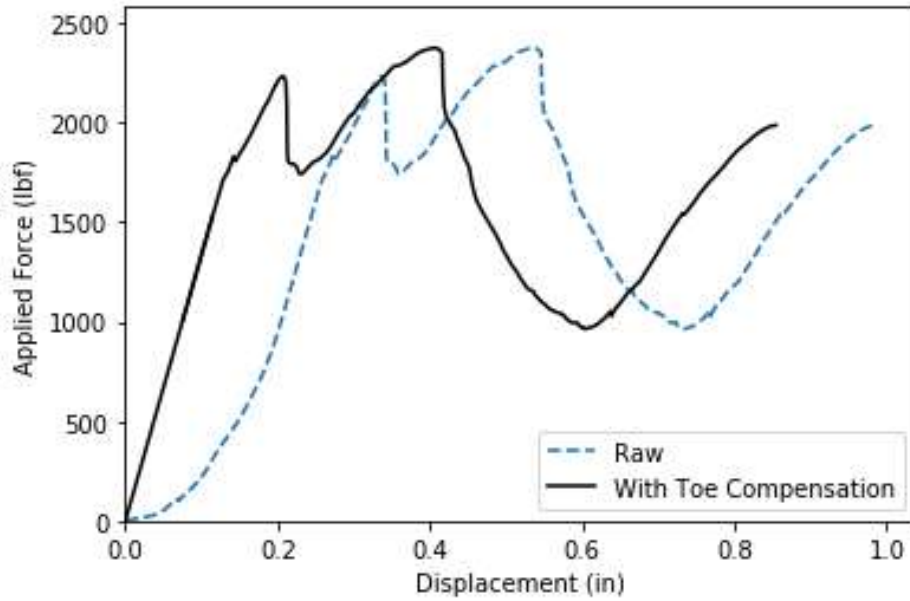


Figure 17: Load-displacement curve for the composite I beam with and without toe compensation

3.4 Isolated Web Test

The first beam assembly with the failed flanges served as a standalone web for additional data collection. The flanges were sawed off, and the resulting piece is shown in Fig. 18.



Figure 18: First beam attempt with flanges removed to isolate the web

The web coupon was placed in the load frame in the same orientation as the web of the beam with flanges. The load-displacement response is shown in Fig. 19. There is a very small dip at 757.7 pounds of applied force, but the reduction in force was only

16 pounds of force. It is unlikely that this was a ply failure since the load shedding is so small. However, at 785.5 pounds of applied force and at a displacement of 0.379 inches, 103.7 pounds dropped as the first ply failed. A more detailed discussion of why this failure occurred will be provided in the failure analysis section.

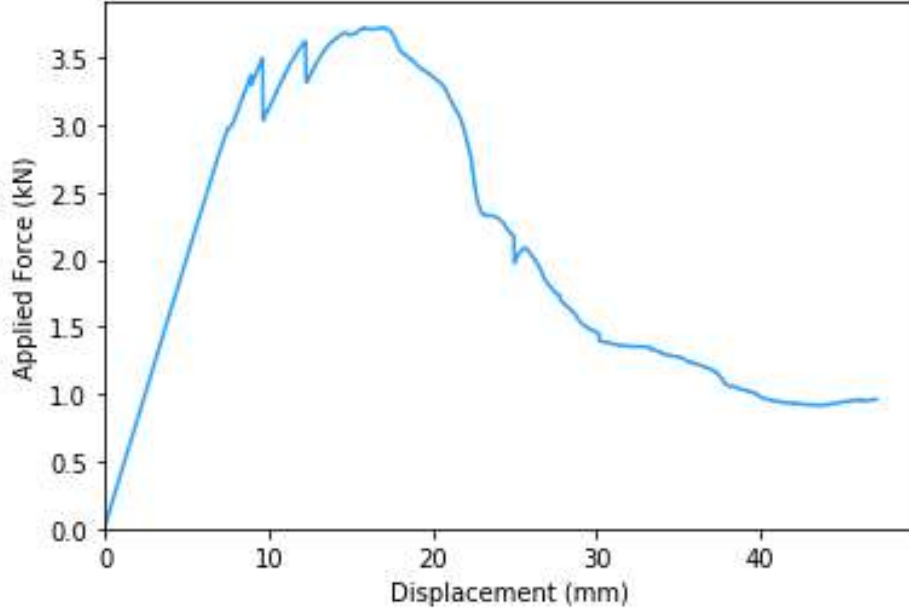


Figure 19: Applied force versus displacement for the isolated web sample

3.5 Calculations

The effective elastic modulus of the beam is determined by identifying the linear elastic region of the load (P in lbf) versus displacement (δ in in/in) curve and performing linear regression to find the slope P/δ . The bending stiffness EI_{eff} is given by Equation 35

$$(EI)_{eff} = \frac{PL^3}{48\delta} \quad (35)$$

where L is the loading span of the beam. The total moment of inertia of the beam cross-section is given by Equation 36

$$\frac{1}{12}(t_{web}(d - 2t_f)^3 + \frac{1}{12}(w(t_f)^3 + A_f \left(\frac{d - t_f}{2}\right)^2) \quad (36)$$

where t_{web} is the measured web thickness, t_f is the measured flange thickness, $A_f = t_f w$ is the cross-sectional area of the flange, and w is the measured width of the flange. Dividing Equation 35 by Equation 36 gives the equivalent flexural modulus E_x of the entire beam.

To determine the failure stress (at the first peak), the beam is represented as a symmetric composite of three parts: web, top flange, and bottom flange. The flanges and web each have their own equivalent elastic modulus. The results of the standalone

web test, described in Section 3.4, are used to estimate the web elastic modulus. Since the processing conditions were different for the web coupon and the final beam, there is expected to be some error related to the laminate material properties (i.e. a higher FVF due to greater temperature and pressure used to cure the standalone web). The equivalent modulus is computed as in the full beam test, using Equation 37

$$(EI)_{web} = \frac{PL^3}{48\delta} \quad (37)$$

where the slope P/δ is derived from the linear elastic region of the load-displacement curve for the standalone web test (Fig. 19, displacement of 0 to 0.08 in). Note that the span, L of the web coupon is equal to the span of the whole beam. The web elastic modulus is computed by dividing the result of Equation 37 by

$$I_{web} = (1/12)(t_{web}(d - 2t_f)^3. \quad (38)$$

From here, the flange elastic modulus is easily computed using the web and whole beam stiffness values in Equation 40.

$$(EI)_{eff} = (EI)_{web} + 2(EI)_{flange} \quad (39)$$

$$E_{flange} = \frac{(EI)_{eff} - (EI)_{web}}{2I_{flange}} \quad (40)$$

An effective cross-section is produced for the web in terms of the ratio n of elastic modulus of the web to that of the flanges. The transformed beam section moment of inertia about the neutral axis is given by Equation 41.

$$I_{tr} = 2I_{flange} + nI_{web} \quad (41)$$

The maximum stress $\sigma_{Mf,flange}$ in the flange and $\sigma_{Mf,web}$ in the web at initial failure point are computed separately using Equations 42 and 43, respectively. P_{m1} is the load corresponding to the first peak in the force-displacement curve. The flange is assumed to be thin, so that the maximum distance from the neutral axis is approximately $d/2$ for both the flange and web.

$$\sigma_{m,flange} = \frac{P_{m1}Ld}{8I_{tr}} \quad (42)$$

$$\sigma_{m,web} = n \frac{P_{m1}Ld}{8I_{tr}} \quad (43)$$

Since the flange equivalent elastic modulus is found to be greater than that of the web, n is less than 1. Therefore, the failure stress in the flange must exceed that of the web. The equation for stress in the web is still useful since the first failure of the physical specimen was observed in the web.

Strain at failure (ε_f) is determined from the maximum deflection offset by the toe compensation value. From the Euler-Bernoulli beam theory, strain in the outer surfaces of a simply-supported beam under a central point load is related to the center point deflection by Equation 44.

$$\varepsilon_f = \frac{6D_{m1}d}{L^2} \quad (44)$$

Here, D_{m1} is the displacement corresponding to the first load peak. Deflection at initial failure (δ_f) is readily extracted from the load-displacement curve. The results of the calculations are summarized in Table 7.

E_x [GPa]	14.63
σ_{fM} (in the flange) [MPa]	115.4
ε_f	0.00704
δ_f [mm]	5.25

Table 7: Summary of values derived from the physical test data

Notably, the value of E_x is approximately 1/3 of the flexural modulus predicted by CLT for the I-beam. This may be attributed to a few factors including reduced fiber volume fraction and increased void volume fraction due to low consolidation pressure, imperfect interlaminar bonding in the web, and crack formation where adjacent parts of plies meet (see Fig. 12).

Fig. 20 shows the beam at peak deflection. Rather than simply forming an arc under load, the ends were crushed, and the majority of the span remained straight. The displacement at failure predicted by CLT (4.9 mm) is comparable to that observed in the physical test, with an error of 6.7%, despite the notable differences in deformed geometry.



Figure 20: I beam in the load frame at at final deflection. Note: this corresponds to the deflection at the end of the test, not the deflection at initial failure.

Calculated stress vs. strain in the flanges and web, including the calculated stress vs. strain for the standalone web are plotted in Fig. 21.

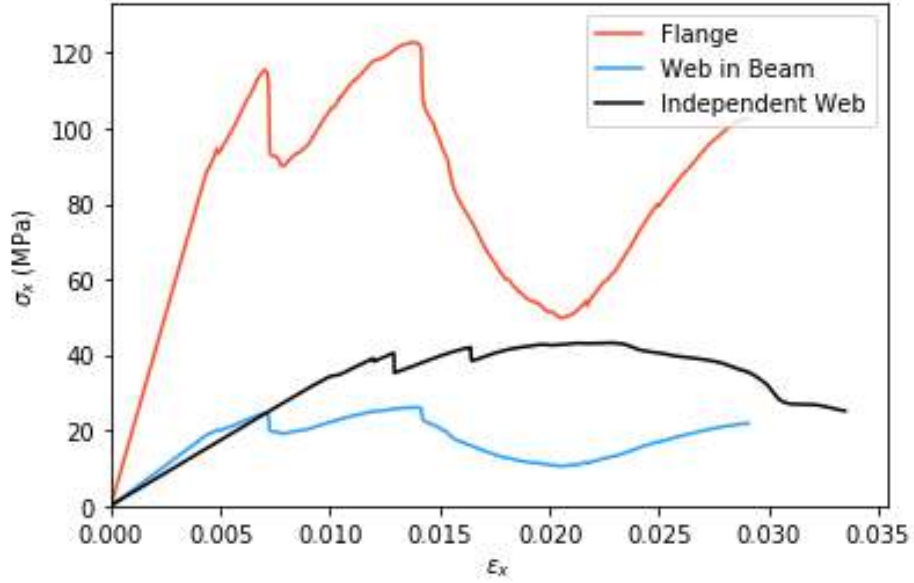


Figure 21: Flexural stress-strain response of the composite I-beam under three-point bending. Stress is calculated at the maximum distance from the neutral axis in the flange and web. The curve obtained from the standalone web test is included for comparison.

4 Failure Analysis

4.1 Failure Documentation

The cause of failure in the web section was clearly visible as soon as the sample was removed from the load frame. In the area around the point at which the failure occurred, four distinct lines can be seen. They are highlighted in fig. 22. These lines are the result of the overlap of the 45° plies, which all occur in the middle of the web. This is also right where the load is applied and due to the overlap, there are no continuous fibers directly underneath where the load is applied. This results in the matrix carrying a significant portion of the load rather than the fibers. As a result, the failure was likely an inter-ply delamination caused by the resin failing. The plies themselves likely survived the failure, resulting in the clean 45° lines observed in the web after failure.

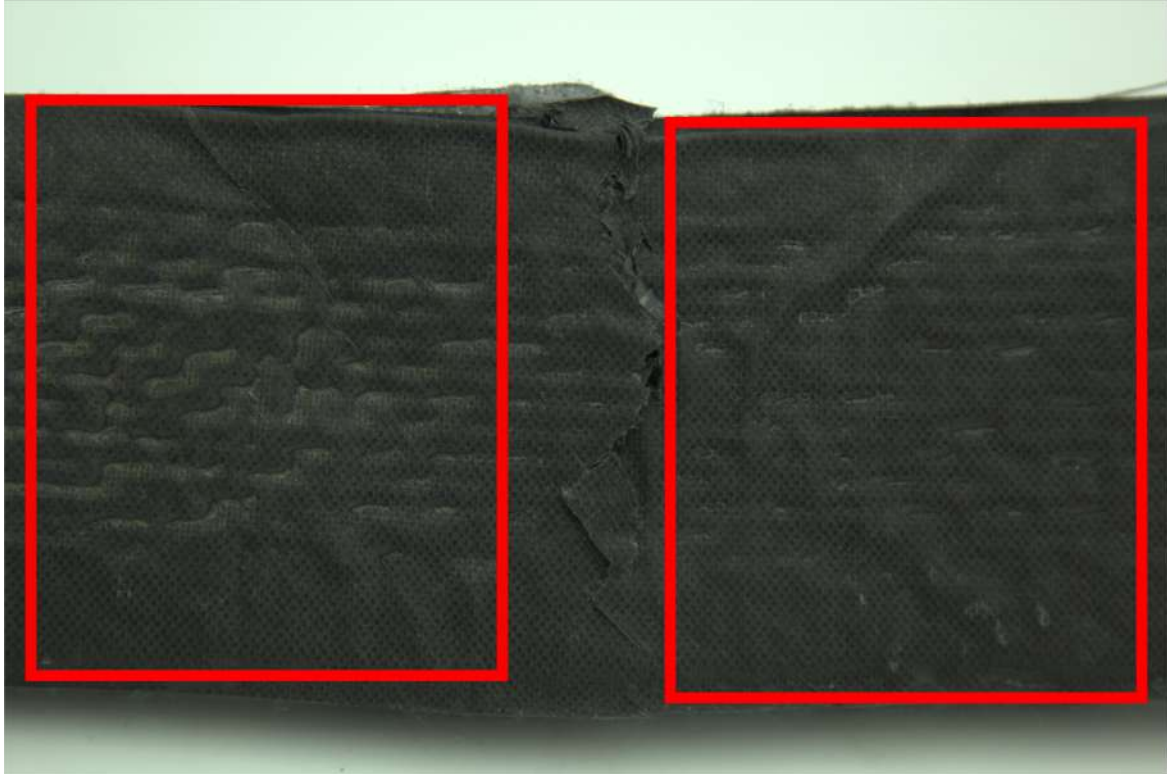


Figure 22: Signs of buckling at the overlap in 45° plies clearly indicated by the clean lines around the failure.

4.2 Stress-Strain Curve to Failure type

Due to the variation in longitudinal elastic modulus for each section of the beam, a single uniform stress-strain curve is not sufficient to describe the response. Thus, Fig. 21 includes calculated stress-strain curves for the web and flanges separately. The curve for the standalone web is included for comparison with the approximate stress response of the integrated web.

Across these regions, the response initially exhibits a linear relationship between stress and strain, corresponding to elastic deformation. The initial nonlinearity in the raw data due to settling is removed for the calculations. The linear region is followed by a slight reduction in stiffness before the peak. This is likely the result of matrix cracking. Next, a sharp drop in stress is observed, indicating fiber failure.

After the initial drop, the beam sections continue to carry load. up to the second peak. The second peak is slightly higher than the first because, after the weakest fibers fail, load is redistributed to stronger fibers. When enough fibers break, the curve falls again. The process then repeats for the final load increase before the test is concluded. This behavior is common for of fiber-reinforced composite materials, where load redistribution occurs after local damage rather than immediate catastrophic failure.

4.3 Variance in Actual Failure Behavior vs. Expected

Theoretical FPF predicted the upper flange would fail in compression first, neglecting failure of the Nomex core. In the physical test, the first signs of damage were observed in the web. Possible reasons for this include cracks forming at the interfaces of adjacent lamina segments, excessive axial load entering the angle-ply of the web, and out-of-plane stresses arising in the web. From Fig. 20, the flanges underwent significant deflection without breaking. As a result, higher loads may have been transferred to the core and web than predicted by the model.

Importantly, the measured peak load from the physical test was less than half the peak load at FPF predicted by the model. Several factors contribute to the observed variance between the actual and expected failure behavior. One primary source is manufacturing variability inherent in composite fabrication. Fiber misalignment can occur during layup, resulting in local deviations from the intended load-carrying directions and reduced effective stiffness and strength. Additionally, defects such as voids and fiber wrinkling introduced during the manufacturing process serve as crack initiation sites. Such material inconsistencies are common in composite specimens and can significantly influence both the location and progression of failure.

5 Discussion and Conclusion

In this report, a CFRP I-beam is designed to undergo 3-point bending to a design load of 7200 lbf. An analytical model using Classical Laminate Theory and Timoshenko Beam Theory is developed, and the predicted first-ply-failure in the upper flange is 7700 lbf. Physical test results showed a peak load at failure of 2376 lbf. Predicted peak deflection is comparable to the model (4.9 vs. 5.25 mm, respectively). The calculated flexural modulus from physical data is roughly 1/3 that predicted by the model (15 and 46 GPa, respectively).

Discrepancies between physical test results and measured values are attributed mainly to defects introduced during the layup and cure process. Additionally, the material properties used for the analysis were not all verified, and some variation is expected. In future projects, a thorough review of the analytical model should be carried out to ensure the prediction includes an adequate safety factor ($\gamma=2.5$).

The sensitivity analysis provides useful practical guidance. Beam height has the largest effect on stiffness, so small geometric inaccuracies can significantly impact performance. In contrast, small fiber angle errors (on the order of a few degrees) have a relatively minor effect, meaning manufacturing effort should prioritize geometry and consolidation over perfect ply alignment.

Experiments in the manufacturing phase revealed some important considerations for future designs. For example, filling the Nomex honeycomb with expanding foam did provide improved stiffness, but required either constraint or better tools to produce a

sufficiently level surface for adhering laminates. Additionally, the first beam iteration demonstrated that autoclave pressure is not generally enough to constrain a part, and physical molding components should be used wherever possible.

The analytical modeling step in Assignment 1 was a valuable experience for learning the strengths and limitations in the design of composite structures. Challenges included defining a reasonable scheme for modeling the 3D beam using 2D CLT analysis, identifying appropriate relations to use for the engineering constants, and ensuring continuity between the virtual beam and the fabrication plan. Since the virtual analysis was used as a basis for the entire beam construction, keeping track of potential errors was crucial. In the future, the process could be facilitated by identifying a collaborative programming environment and having more discussions about the modeling process.

Another important insight from this study is the strong coupling between manufacturing quality and structural performance in the composites. The analytical model assumes ideal conditions such as uniform fiber distribution and perfect bonding. However, the experimental results highlight the structure's sensitivity to imperfections such as voids and poor adhesion. These defects act as stress concentrators and lead to the premature failure well below the predicted load. Additionally, the reduced flexural modulus indicates that the stiffness is influenced by both geometric variations and inefficient load transfer across the laminate. Overall, the progress from design to fabrication and testing significantly improved the understanding of composite structures as an integrated process. The ability to directly compare predicted and experimental results helped identify gaps between theory and practice.

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